



Commonwealth of Virginia

VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

www.deq.virginia.gov

Stefanie K. Taillon
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus
Director

November 6, 2025

Mr. Chris Napier
Senior Vice President of Operations
CyrusOne, LLC
2850 N Harwood St, Suite 2200
Dallas, TX 75201

Location: Loudoun County
Registration No.: 74173

Dear Mr. Napier:

Attached is a permit document to construct and operate diesel engine generator sets (gen-sets) at the CyrusOne Sterling NVA9-NVA11 data center in accordance with the provisions of the Virginia State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. This permit document supersedes your permit document dated November 19, 2024.

In the course of evaluating the application and arriving at a final decision to approve the project, the Department of Environmental Quality (DEQ) deemed the application complete on November 3, 2025.

This permit contains legally enforceable conditions. Failure to comply may result in a Notice of Violation and/or civil charges. Please read all permit conditions carefully.

This permit approval to construct and operate shall not relieve CyrusOne, LLC of the responsibility to comply with all other local, state, and federal permit regulations.

The proposed diesel engine gen-sets may be subject to the requirements of 40 CFR Part 60, New Source Performance Standards (NSPS) Subpart IIII – Standards of Performance for Stationary Compression Ignition Internal Combustion Engines and 40 CFR Part 63, National Emission Standards for Hazardous Air Pollutants (MACT) Subpart ZZZZ – National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines. In summary, the units may be required to comply with certain federal emission standards and operating limitations. The Department of Environmental Quality (DEQ) advises you to review the referenced MACT and NSPS to ensure compliance with applicable emission and operational limitations. As the owner/operator you are also responsible for any monitoring, notification, reporting and recordkeeping requirements of the MACT and NSPS. Notifications shall only be sent to EPA, Region III.

To review any federal rules referenced in the above paragraph or in the attached permit, the US Government Publishing Office maintains the text of these rules at www.ecfr.gov, Title 40, Part 60 and 63.

The Board's Regulations as contained in Title 9 of the Virginia Administrative Code 5-170-200 provide that you may request a formal hearing from this case decision by filing a petition with the Board within 30 days after this case decision notice was mailed or delivered to you. Please consult the relevant regulations for additional requirements for such requests.

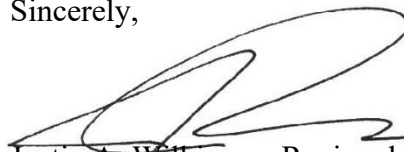
As provided by Rule 2A:2 of the Supreme Court of Virginia, you have 30 days from the date you actually received this permit or the date on which it was mailed to you, whichever occurred first, within which to initiate an appeal of this decision by filing a Notice of Appeal with:

Michael S. Rolband, Director
Department of Environmental Quality
P. O. Box 1105
Richmond, VA 23218

If this permit was delivered to you by mail, three days are added to the thirty-day period in which to file an appeal. Please refer to Part Two A of the Rules of the Supreme Court of Virginia for information on the required content of the Notice of Appeal and for additional requirements governing appeals from decisions of administrative agencies.

If you have any questions concerning this permit, please contact Ms. Katie DeVoss at (571) 866-6090 or katie.devoss@deq.virginia.gov.

Sincerely,



Justin A. Wilkinson, Regional Air Permit Manager
Virginia Department of Environmental Quality
justin.wilkinson@deq.virginia.gov
Northern Regional Office
13901 Crown Court, Woodbridge, VA 22193
(703) 583-3800

JAW/KD/74173 mNSR (2025-11-06)
Attachment: Permit



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VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

www.deq.virginia.gov

Stefanie K. Taillon
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus
Director

STATIONARY SOURCE PERMIT TO CONSTRUCT AND OPERATE

This permit document supersedes your permit document dated November 19, 2024.

In compliance with the Federal Clean Air Act and the Commonwealth of Virginia Regulations for the Control and Abatement of Air Pollution,

CyrusOne, LLC
2850 N Harwood Street, Suite 2200
Dallas, Texas 75201
Registration No.: 74173

is authorized to construct and operate

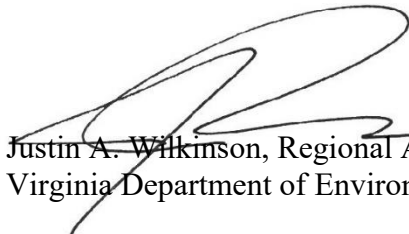
diesel engine generator sets (gen-sets)

located at

CyrusOne NVA9-NVA11 Facility
43290 Old Ox Road
Sterling, Virginia 20166
(Loudoun County)

in accordance with the Conditions of this permit.

Approved on: November 6, 2025.


Justin A. Wilkinson, Regional Air Permit Manager
Virginia Department of Environmental Quality

Permit consists of 24 pages.
Permit Conditions 1 to 34.

INTRODUCTION

This permit approval is based on and combines permit terms and conditions in accordance with 9VAC5-80-1255 from the following permit approvals and the respective permit applications:

- mNSR permit dated November 6, 2025, based on the permit application dated March 11, 2025, including supplemental information dated April 19, 2025, May 21, 2025, July 3, 2025, September 22, 2025, and November 3, 2025.
- mNSR amendment dated November 19, 2024, based on the permit application dated October 14, 2024;
- mNSR permit dated March 15, 2023, based on the permit application dated February 10, 2023, including supplemental information dated March 2, 2023;
- mNSR permit dated December 19, 2022, based on the permit application dated October 28, 2021, including supplemental information dated March 24, 2022, October 14, 2022, and November 22, 2022;
- mNSR permit dated June 3, 2021, based on the permit application dated April 16, 2021, including supplemental information dated May 6, 2021;
- mNSR permit dated November 5, 2019, based on the permit application dated March 14, 2019, including supplemental information dated May 22, 2019, October 7, 2019, October 10, 2019, and November 1, 2019.

Any changes in the permit application specifications or any existing facilities, which alter the impact of the facility on air quality, may require a permit. Failure to obtain such a permit prior to construction may result in enforcement action. In addition, this facility may be subject to additional applicable requirements not listed in this permit.

Words or terms used in this permit shall have meanings as provided in 9VAC5-80-1110 and 9VAC5-10-10 of the Commonwealth of Virginia's State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. The regulatory reference or authority for each condition is listed in parentheses () after each condition. The enabling permit program, or permit programs is provided below each permit condition in the regulatory authority parenthetical as follows: 9VAC5-80-850 for Article 5, 9VAC5-80-1180 for Article 6, 9VAC5-80-1985 for Article 8, and 9VAC5-80-2050 for Article 9. The most recent effective date for a term or condition is listed in brackets []. When identical conditions for one or more emission units are combined, the effective date listed in this permit does not alter the prior effective date(s) for any such conditions as issued in a previous permit action. In accordance with 9VAC5-80-1120F, any condition not marked as state-only enforceable (SOE) is state and federally enforceable.

Annual requirements to fulfill legal obligations to maintain current stationary source emissions data will necessitate a prompt response by the permittee to requests by the Department of Environmental Quality (DEQ) or the Board for information to include, as appropriate: process and production data; changes in control equipment; and operating schedules. Such requests for information from the DEQ will either be in writing or by personal contact.

The availability of information submitted to the DEQ or the Board will be governed by applicable provisions of the Freedom of Information Act, §§ 2.2-3700 through 2.2-3714 of the Code of Virginia, § 10.1-1314 (addressing information provided to the Board) of the Code of Virginia, and 9VAC5-170-60 of the State Air Pollution Control Board's Regulations. Information provided to federal officials is subject to appropriate federal law and regulations governing confidentiality of such information.

Equipment List – Equipment at this facility consists of:

Equipment to be Constructed:

Engine Group	Reference No.	Equipment Description	Rated Capacity (each unit)	Add-On Control Technology	Delegated Federal Requirements	Original Permit Date
Group 3C	NVA10_01 through NVA10_28, NVA10_House	Twenty-nine (29) non-emergency diesel engine gen-sets consisting of: Cummins QSK60-G19 (Generator Model DQKAN) and/or MTU 16V4000G84S (Generator Model 16V4000 DS2250)	Cummins: 2,250 ekW 3,202 bhp MTU: 2,250 ekW 3,353 bhp	Selective Catalytic Reduction (SCR) and catalyzed Diesel Particulate Filter (cDPF)*	None	November 6, 2025
Group 4	NVA10_29 through NVA10_32, NVA11_01 through NVA11_15	Nineteen (19) non-emergency diesel engine gen-sets consisting of: Cummins QSK60-G19 (Generator Model DQKAN) and/or MTU 16V4000G84S (Generator Model 16V4000 DS2250)	Cummins: 2,250 ekW 3,202 bhp MTU: 2,250 ekW 3,353 bhp	SCR and cDPF*	None	November 6, 2025
Group 5	NVA11_House	One (1) Cummins QSK38-G17 emergency diesel engine gen-set (Generator Model C1250DE)	Cummins: 1,250 ekW 1872 bhp		None	November 6, 2025

* Miratech (Model No. M3BZ-64-56-J-TBD-R4, ACIS-3) for any Cummins unit and a Catalytic Combustion system (Model No. 1050066) for any MTU unit.

Equipment Previously Permitted:

Engine Group	Reference Nos.	Equipment Description	Standby Rated Capacity (each unit)	Delegated Federal Requirements	Original Permit Date
Group 1	101-GEN-A1 through 101-GEN-D1, 101-GEN-A2 through 101-GEN-D2, 102-GEN-A1 through 102-GEN-D1, 201-GEN-A1 through 201-GEN-D1	Sixteen (16) MTU 16V4000G84S emergency diesel engine gen-sets (Generator Model 16V4000 DS2250)	3,353 bhp 2,250 ekW	None	November 5, 2019
House Gen	H-GEN-1A	One (1) MTU 16V4000G74S emergency diesel engine gen-set (Generator Model 16V4000 DS2000)	3,058 bhp 2,000 ekW	None	November 5, 2019
Group 2A	201-GEN-A2 and 201-GEN-R2	Two (2) Cummins QSK60-G19 emergency diesel engine gen-sets (Generator Model DQKAN)	3,202 bhp 2,250 ekW	None	June 3, 2021
Group 2B	201-GEN-B2, 201-GEN-C2, 202-GEN-A1 through 202-GEN-F1, 202-GEN-R1	Nine (9) Cummins QSK60-G19 emergency diesel engine gen-sets (Generator Model DQKAN)	3,202 bhp 2,250 ekW	None	December 19, 2022
Group 3A	202-GEN-G1	One (1) CAT 3516C emergency diesel engine gen-set	3,634 bhp 2,500 ekW	None	March 15, 2023
Group 3B	102-GEN-A2, 102-GEN-B2, 102-GEN-C2, and 102-GEN-R2	Four (4) Cummins QSK60-G19 emergency diesel engine gen-sets (Generator Model DQKAN)	3,202 bhp 2,250 ekW	None	December 19, 2022

Specifications included in the above tables are for informational purposes only and do not form enforceable terms or conditions of the permit.

PROCESS REQUIREMENTS

1. **Emission Controls (SCR)** – Nitrogen oxide (NO_x) emissions from the non-emergency diesel engine gen-sets (Engine Groups 3C and 4) shall be controlled by closed loop Selective Catalytic Reduction (SCR). Each SCR system shall be equipped with a temperature probe to continuously monitor the catalyst bed exhaust temperature while the engine gen-set is operational.
 - a. For the Cummins QSK60-G19 units:
 - i. Engine exhaust gas shall be treated with urea when the catalyst bed exhaust temperature of 572°F is achieved.
 - ii. The permittee shall operate the engine gen-set and SCR such that the catalyst bed exhaust temperature does not exceed 977°F.
 - b. For the MTU 16V4000G84S units:
 - i. Engine exhaust gas shall be treated with urea when the catalyst bed exhaust temperature of 500°F is achieved.
 - ii. The permittee shall operate the engine gen-set and SCR such that the catalyst bed exhaust temperature does not exceed 977°F.

The SCR shall be provided with adequate access for inspection and shall be in operation when the engine gen-sets are operating as stated above.
(9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

2. **Emission Controls (cDPF)** – Particulate matter (PM, PM₁₀, and PM_{2.5}), carbon monoxide (CO), and volatile organic compounds (VOC) emissions from the non-emergency diesel engine gen-sets (Engine Groups 3C and 4) shall be controlled by a catalyzed Diesel Particulate Filter (cDPF). The cDPF shall be provided with adequate access for inspection and shall be in operation when the non-emergency diesel engine gen-sets are operating.
(9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

3. **Emission Controls –**

- a. Nitrogen oxide (NO_x) emissions from each emergency diesel engine gen-set (Engine Groups 1, 2A, and House Gen) shall be controlled by electronic fuel injection, turbocharged engines, and charge air coolers. The permittee shall maintain documentation that demonstrates the control devices have been installed on each emergency diesel engine gen-set.
- b. Nitrogen oxide (NO_x) emissions from each emergency diesel engine gen-set (Engine Groups 2B, 3A, 3B, and 5) shall be controlled by engine design.

(9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

4. **Emission Controls –** Visible emissions, particulate matter (PM₁₀ and PM_{2.5}) emissions, carbon monoxide (CO) emissions, volatile organic compound (VOC) emissions, and nitrogen oxide (NO_x) emissions from the emergency diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen) shall be controlled by the use of good operating practices and performing appropriate maintenance in accordance with the manufacturer recommendations. In addition, the permittee may only change those settings that are permitted by the manufacturer and do not increase air emissions.
(9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

5. **Monitoring Devices –**

- a. Fuel Flow: Each diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) shall be equipped with a device to continuously measure and record individual fuel consumption (in gallons) for each engine gen-set.
- b. Engine Operating Hours: Each diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) shall be equipped with a non-resettable hour metering device to monitor the operating hours.

Each monitoring device (as required in a. and b. above) shall be observed by the owner with a frequency of not less than once each day the emergency diesel engine gen-set is operated. The owner shall keep a log of these observations.

Each monitoring device shall be installed, maintained, calibrated (as appropriate) and operated in accordance with approved procedures, which shall include, as a minimum, the manufacturer's written requirements or recommendations. The details of the monitoring device calibrations are to be arranged with the Regional Air Compliance Manager of the DEQ's Northern Regional Office (NRO).

Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the emergency diesel engine gen-sets are operating.
(9VAC5-80-1180 D) [11/6/2025]

6. **Monitoring Devices (SCR and cDPF)** – Each non-emergency diesel engine gen-set (Engine Groups 3C and 4) shall be equipped with devices to continuously measure and record the following:
- a. The SCR catalyst bed exhaust temperature. The information shall be recorded at a minimum frequency of once every fifteen minutes, and correlated to run date, engine operating hours, and fuel consumption.
 - b. The NO_x concentration measured after the SCR catalyst, expressed in ppm. The information shall be recorded at a minimum frequency of once every fifteen minutes and correlated to run date, engine operating hours, and fuel consumption.
 - c. The differential pressure drop across the cDPF to demonstrate proper function. The backpressure monitor shall notify the permittee when the high backpressure limit of the engine is approached. The differential pressure drop shall be recorded by the permittee with a frequency as recommended by the process/control equipment manufacturer.
 - d. The cDPF catalyst bed exhaust temperature. The information shall be recorded at a minimum frequency of once every fifteen minutes and correlated to run date, engine operating hours, and fuel consumption.

Each monitoring device shall be installed, maintained, calibrated and operated in accordance with approved procedures which shall include, as a minimum, the manufacturer's written requirements or recommendations. Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the SCR and cDPF are operating. (9VAC5-80-1180 D) [11/6/2025]

OPERATING LIMITATIONS

7. **Operation of the Engine Gen-Sets** – The permittee shall operate and maintain each diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) and control device according to the manufacturer's written instructions or procedures developed by the permittee that are approved by the engine manufacturer. In addition, the permittee may only change those settings that are permitted by the manufacturer and do not increase air emissions. (9VAC5-80-1180) [11/6/2025]
8. **Operating Limitations (Ozone Season)** – No diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) shall be operated for scheduled maintenance, readiness testing, or operation training (that involves fuel combustion) between the hours of 7 AM to 5 PM any day during the ozone season of May 1 through September 30. The permittee may petition the Regional Air Compliance Manager of the DEQ's NRO for exceptions to this requirement, with approvals made on a case by case basis. (9VAC5-80-1180) [11/6/2025]

9. **Operating Limitations (Ozone Season) - Integration Operational Period** – During the integration operational period of each diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen), any operation of the unit (that involves fuel combustion) between the hours of 7 AM to 5 PM any day during the ozone season of May 1 through September 30 shall only occur if the forecast Air Quality Index (AQI) for ozone as published on the AirNow website (<http://airnow.gov>) for Northern Virginia for that day is ≤ 100 . In the event that AirNow-EnviroFlash (www.enviroflash.info) issues an Air Alert for Metropolitan Washington, D.C. for a day which the forecasted AQI for ozone was ≤ 100 , operation of each unit (which involves fuel combustion) shall be minimized to the maximum extent practical. (9VAC5-80-1180) [11/6/2025]
10. **Emergency Power Generation** – The emergency diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen) shall only be operated in the following modes:
 - a. In situations that arise from sudden and reasonably unforeseeable events where the primary energy or power source is disrupted or disconnected due to conditions beyond the control of an owner or operator of a facility including:
 - i. A failure of the electrical grid;
 - ii. On-site disaster or equipment failure; or
 - iii. Public service emergencies such as flood, fire, natural disaster, or severe weather conditions.
 - b. An Independent System Operator (ISO) declared emergency, where an ISO emergency is any of the following:
 - i. An abnormal system condition requiring manual or automatic action to maintain system frequency, to prevent loss of firm load, equipment damage, or tripping of system elements that could adversely affect the reliability of an electric system or the safety of persons or property;
 - ii. Capacity deficiency or capacity excess conditions;
 - iii. A fuel shortage requiring departure from normal operating procedures in order to minimize the use of such scarce fuel;
 - iv. Abnormal natural events or man-made threats that would require conservative operations to posture the system in a more reliable state; or
 - v. An abnormal event external to the ISO service territory that may require ISO action.
 - c. For scheduled maintenance checks and readiness testing (Scheduled MCRT).

- d. For unscheduled maintenance, testing, and operational training.
- e. For the integration operational period, which is the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the sources electrical system.

(9VAC5-80-1180) [11/6/2025]

11. **Operating Hours** – The operating hours of the emergency diesel engine gen-sets are limited to the following:

- a. No single unit (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen) shall operate more than 500 hours per year for all purposes (as provided in Condition 10), calculated monthly as the sum of each consecutive 12-month period;
- b. No single unit (Engine Groups 1, 2A, 2B, 3A, 3B, and House Gen) shall operate more than 18 hours per year for scheduled maintenance checks and readiness testing (Scheduled MCRT, as provided in Condition 10.c).
- c. No single unit (Engine Group 5) shall operate more than 22 hours per year for scheduled maintenance checks and readiness testing (Scheduled MCRT, as provided in Condition 10.c).

Compliance for each consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.

(9VAC5-80-1180) [11/6/2025]

12. **Fuel Specification** – The approved fuel for the diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) is ultra-low sulfur diesel fuel oil and shall meet the specifications below:

ULTRA-LOW SULFUR DIESEL FUEL OIL:

- a. Does not exceed the American Society for Testing and Materials (ASTM) specification, D975, for grade ultra-low sulfur 1-D or grade 2-D S15, or
- b. Has a maximum sulfur content not to exceed 0.0015% by weight (15 ppm), and either a minimum cetane number of 40 or maximum aromatic content of 35 volume percent.

Exceedance of these specifications may be considered credible evidence of an exceedance of emission limits. A change in the fuel type or the fuel sulfur content may require a permit to modify and operate.

(9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

13. **Fuel Certification** – The permittee shall obtain a certification from the fuel supplier with each shipment of diesel fuel oil. Each fuel supplier certification shall include the following:
- a. The name of the fuel supplier;
 - b. The date on which the diesel fuel oil was received;
 - c. The quantity of diesel fuel oil delivered in the shipment;
 - d. A statement that the diesel fuel oil:
 - i. complies with the ASTM specifications for Grade No. 1-D S15 or Grade No. 2-D S15 (also known as ultra low sulfur diesel (ULSD)); or
 - ii. has a sulfur content per shipment not to exceed 0.0015% by weight (15ppm) and either a minimum cetane number of forty or maximum aromatic content of thirty-five percent by volume.

Alternatively, the permittee must obtain approval from the Regional Air Compliance Manager of the DEQ's NRO, if other documentation will be used to certify the diesel fuel oil type.

Fuel sampling and analysis, independent of that used for certification, as may be periodically required or conducted by DEQ may be used to determine compliance with the fuel specifications stipulated in Condition 12. Exceedance of these specifications may be considered credible evidence of the exceedance of emission limits.
(9VAC5-80-1180) [11/6/2025]

14. **Fuel Throughputs** – The diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) for all operations shall only consume a combined quantity of diesel fuel (in gallons) each consecutive 365-day period, as demonstrated by using the following equation:

$$\frac{\text{Group A Fuel Consumption}}{4,731,618 \text{ gallons}} + \frac{\text{Group B Fuel Consumption}}{521,562 \text{ gallons}} + \frac{\text{Group C Fuel Consumption}}{673,204 \text{ gallons}} \leq 1$$

Where:

Fuel Group	Engine Groups	Engine Manufacturer	SCR and cDPF Status
Group A	Groups 3C and 4	MTU & Cummins	SCR and cDPF Controlled
Group B	Groups 3C and 4	MTU Only	SCR Uncontrolled and cDPF Controlled/Uncontrolled
Group C	Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen	MTU, Cummins, & Caterpillar	N/A
	Groups 3C and 4	Cummins Only	SCR Uncontrolled and cDPF Controlled/Uncontrolled

Compliance for the consecutive 365-day period shall be demonstrated daily by adding the total for the most recently completed calendar day to the individual daily total for the preceding 364 days.

(9VAC5-80-1180) [11/6/2025]

EMISSION LIMITS

15. **Emission Limits (Hourly - Emergency)** – Emissions from the operation of each of the emergency diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen) shall not exceed the limits specified below:

Pollutant	Engine Group 1	Engine Groups 2A, 2B, and 3B	Engine Group 3A
Nitrogen Oxides (as NO ₂)	44.28 lbs/hr	42.36 lbs/hr	48.06 lbs/hr
Carbon Monoxide (CO)	9.36 lbs/hr	9.88 lbs/hr	6.01 lbs/hr
Volatile Organic Compounds (VOC)	2.20 lbs/hr	0.56 lbs/hr	1.20 lbs/hr
Particulate Matter (PM)	0.78 lbs/hr	0.88 lbs/hr	0.41 lbs/hr
Particulate Matter (PM ₁₀)	0.91 lbs/hr	1.05 lbs/hr	0.59 lbs/hr
Particulate Matter (PM _{2.5})	0.91 lbs/hr	1.05 lbs/hr	0.59 lbs/hr

Pollutant	Engine Group 5	Engine Group House Gen
Nitrogen Oxides (as NO ₂)	24.76 lbs/hr	40.38 lbs/hr
Carbon Monoxide (CO)	4.99 lbs/hr	7.03 lbs/hr
Volatile Organic Compounds (VOC)	0.32 lbs/hr	2.06 lbs/hr
Particulate Matter (PM)	0.44 lbs/hr	0.91 lbs/hr
Particulate Matter (PM ₁₀)	0.51 lbs/hr	1.03 lbs/hr
Particulate Matter (PM _{2.5})	0.51 lbs/hr	1.03 lbs/hr

Compliance with these emission limits shall be based on the proper operation and maintenance of the emergency diesel engine gen-sets or by testing, if required.
 (9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

16. **Emission Limits (Hourly - Non-Emergency)** – Emissions from the operation of each of the emergency diesel engine gen-sets (Engine Groups 3C and 4) shall not exceed the limits specified below:

Pollutant	Cummins QSK60-G19		MTU 16V4000G84S	
	Controlled*	Uncontrolled	Controlled*	Uncontrolled
Nitrogen Oxides (as NO ₂)	3.53 lbs/hr	42.36 lbs/hr	3.70 lbs/hr	44.35 lbs/hr
Carbon Monoxide (CO)	1.76 lbs/hr	9.88 lbs/hr	1.78 lbs/hr	11.87 lbs/hr
Volatile Organic Compounds (VOC)	0.28 lbs/hr	0.56 lbs/hr	0.97 lbs/hr	2.57 lbs/hr
Particulate Matter (PM)	0.15 lbs/hr	0.88 lbs/hr	0.16 lbs/hr	0.93 lbs/hr
Particulate Matter (PM ₁₀)	0.31 lbs/hr	1.05 lbs/hr	0.33 lbs/hr	0.96 lbs/hr
Particulate Matter (PM _{2.5})	0.31 lbs/hr	1.05 lbs/hr	0.33 lbs/hr	0.96 lbs/hr

* Exhaust gas treated with both SCR and cDPF

Compliance with these emission limits shall be based on the proper operation and maintenance of the emergency diesel engine gen-sets, pollution control devices, or by testing, if required.

(9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

17. **Emission Limits (Annual)** – Total emissions from the combined operation of the emergency diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) for all purposes shall not exceed the limits specified below:

Pollutant	Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen
Nitrogen Oxides (as NO ₂)	94.43 tpy
Carbon Monoxide (CO)	50.33 tpy
Volatile Organic Compounds (VOC)	46.84 tpy
Particulate Matter (PM)	9.01 tpy
Particulate Matter (PM ₁₀)	9.29 tpy
Particulate Matter (PM _{2.5})	9.29 tpy

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with these emission limits may be determined as stated in Conditions 1, 2, 3, 4, 12, 14, 15, and 16.

(9VAC5-80-1180) [11/6/2025]

18. Visible Emission Limit –

- a. Visible emissions from each non-emergency engine gen-set (Engine Groups 3C and 4) shall not exceed 5% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A). This requirement applies at all times except during startup, shutdown, and malfunction.
- b. Visible emissions from each emergency engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen) shall not exceed 5% opacity except during one six-minute period in any one hour in which visible emissions shall not exceed 10% opacity as determined by the EPA Method 9 (reference 40 CFR 60, Appendix A). This requirement applies at all times except during startup, shutdown, and malfunction.

During startup and shutdown, visible emissions from each engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) shall not exceed 10% opacity except during one six-minute period in any one-hour in which visible emissions shall not exceed 20% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A).
(9VAC5-80-1180 and 9VAC5-50-260) [11/6/2025]

INITIAL COMPLIANCE DETERMINATION

- 19. Stack Test (Groups 3C and 4) –** Initial performance tests shall be conducted on at least twenty percent (20%) of the units installed for each type of engine model (Cummins QSK60-G19 and MTU 16V4000G84S) of the forty-eight (48) non-emergency diesel engine gen-sets (Engine Groups 3C and 4) for controlled NO_x (as NO₂) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ's NRO to determine compliance with the emission limits contained in Condition 16.

- a. Emissions testing of each pollutant for each selected non-emergency diesel engine gen-set shall consist of three (3) one-hour test runs under load. The average of the three (3) runs shall be reported as the short-term emission rate for that non-emergency diesel engine gen-set;
- b. Testing shall be performed on the exhaust stack of the non-emergency diesel engine gen-sets to demonstrate compliance with the controlled NO_x and CO emission limits specified in Condition 16. Testing shall be conducted with the non-emergency diesel engine gen-set operating at ≥ 90 percent of its rated capacity, unless multiple load band testing is approved by DEQ;
- c. Recorded non-emergency diesel engine gen-set operational information shall include, but not be limited to:
 - i. Generator load/kilowatt output.

- ii. Fuel consumption and fuel sulfur content of the diesel fuel oil.
 - iii. SCR catalyst bed exhaust temperature.
 - iv. Urea solution injection rate.
 - v. NO_x concentration after the catalyst.
 - vi. Differential pressure drop across the cDPF.
 - vii. cDPF catalyst bed exhaust temperature.
- d. Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced. The integration operational period is defined as: the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source electrical system. If this deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
- e. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ's NRO, at least 30 days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less than 30 days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;
- f. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ's NRO, in writing, within seven (7) days of the scheduled test date or as soon as the rescheduling is deemed necessary; and
- g. Two (2) copies (one (1) paper copy and one (1) electronic copy) of the test results shall be submitted to the Regional Air Compliance Manager, DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9VAC5-50-30 and 9VAC5-80-1200) [11/6/2025]

20. **Visible Emissions Evaluation** – Concurrent with the initial compliance determinations required in Condition 19, Visible Emission Evaluations (VEE) in accordance with 40 CFR Part 60, Appendix A, Method 9, shall also be conducted by the permittee on the non-emergency diesel engine gen-sets selected for initial performance testing. The details of the tests shall be arranged with the Regional Air Compliance Manager of the DEQ's NRO. The permittee shall submit a VEE protocol in conjunction with the initial stack test protocol required by Condition 19, at least 30 days prior to testing.
- a. Should conditions prevent concurrent opacity observations, the Regional Air Compliance Manager of the DEQ's NRO shall be notified in writing, within seven days, and visible emissions testing shall be rescheduled within 30 days. Rescheduled testing shall be conducted under the same conditions (as possible) as the initial performance tests.
 - b. Two copies of the test result (one hard copy and one on electronic media) shall be submitted to the Regional Air Compliance Manager of the DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit (Attachment A).

(9VAC5-50-30 and 9VAC5-80-1200) [11/6/2025]

CONTINUING COMPLIANCE DETERMINATION

21. **Facility Construction** – The diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) shall be constructed so as to allow for emissions testing upon reasonable notice at any time, using appropriate methods. This includes constructing the facility/equipment such that volumetric flow rates and pollutant emission rates can be accurately determined by applicable test methods and providing a stack or duct that is free from cyclonic flow. Sampling ports shall be provided when requested at the appropriate locations and safe sampling platforms and access shall be provided.
(9VAC5-50-30 F and 9VAC5-80-1180) [11/6/2025]
22. **Emission Testing/Visible Emissions Evaluation** – Upon request by the DEQ, the permittee shall conduct stack tests and/or VEEs of the diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) to demonstrate compliance with the emission limits contained in this permit. The details of the tests shall be arranged with the Regional Air Compliance Manager of the DEQ's NRO.
(9VAC5-80-1200 and 9VAC5-50-30 G) [11/6/2025]

RECORDS

23. **On Site Records** – The permittee shall maintain records of emission data and operating parameters as necessary to demonstrate compliance with this permit. The content and format of such records shall be arranged with the Regional Air Compliance Manager of the DEQ’s NRO. These records shall include, but are not limited to:

- a. A log of the monitoring device observations as required in Condition 5.
- b. Operation and control device monitoring records for each non-emergency engine gen-set equipped with SCR (Engine Groups 3C and 4), as required in Conditions 6.a and 6.b. This includes records of the SCR catalyst bed exhaust temperature and NO_x emission concentration.
- c. Operation and control device monitoring records for each non-emergency engine gen-set equipped with a cDPF (Engine Groups 3C and 4), as required in Conditions 6.c and 6.d. This includes records of the differential pressure drop across the filter and catalyst bed exhaust temperature.
- d. Records for diesel engine gen-set operations, as necessary, to demonstrate compliance with the operating limitations of Condition 8; which includes but is not limited to: times, dates, and reasons for operation of each diesel engine gen-set that was operating between May 1 and September 30.
- e. To verify compliance with Condition 9, maintain records of:
 - i. The forecasted AQI, as determined by the AirNow website for Northern Virginia, for ozone for the day(s) that a diesel engine gen-set operated during the integration operational period;
 - ii. The measured AQI, as determined by the AirNow website for Northern Virginia, for ozone for the day(s) that the diesel engine gen-set operated during the integrational operational period;
 - iii. Documentation recording any Air Alerts issued for the operating day, as determined by AirNow-EnviroFlash; and
 - iv. Details of commissioning activities, to include, but not limited to, clock hours and duration.
- f. Records of the reasons for operation for each emergency diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen), including, but not limited to, the date, cause of operation, cause of the emergency, the ISO-declared emergency notification, and the hours of operation.

- g. Annual hours of operation of each emergency diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen), calculated monthly as the sum of each consecutive 12-month period, as specified in Condition 11.a.
- h. Monthly and annual hours of operation of each emergency diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 5, and House Gen) for purposes of Scheduled maintenance checks and readiness testing (Scheduled MCRT, as provided in Conditions 11.b and 11.c), calculated monthly as the sum of each consecutive 12-month period.
- i. All fuel supplier certifications.
- j. Daily and annual fuel consumption of each diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen), for all purposes, with the annual fuel consumption calculated daily as the sum of each consecutive 365-day period.
- k. Daily and annual fuel consumption for each group of nine (9) diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) by fuel throughput groups (Group A, Group B, and Group C), calculated daily as the sum of each consecutive 365-day period for the purposes of the compliance demonstration with the diesel fuel throughput limit equation in Condition 14.
- l. Daily and annual results of the computed fuel throughput equation used by the groups of nine (9) diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) by fuel throughput groups (Group A, Group B, and Group C), calculated daily as the sum of each consecutive 365-day period for the purposes of the compliance demonstration with the diesel fuel throughput limit equation in Condition 14.
- m. Daily and annual emissions calculations for NO_x (as NO₂), CO, VOC, PM, PM₁₀, and PM_{2.5} from the diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen), with annual emissions calculated daily, as the sum of each consecutive 365-day period, to verify compliance with the annual emission limits in Condition 17.
- n. Documentation from the manufacturer that each diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) is certified to meet the EPA Tier 2 emission standards.
- o. Engine information including make, model, serial number, model year, maximum engine power (bhp), and engine displacement for each diesel engine gen-set.
- p. The manufacturer's written operating instructions or procedures developed by the owner/operator that are approved by the engine manufacturer for each diesel engine gen-set.

- q. Records of changes in settings that are permitted by the manufacturer of the diesel engine gen-sets.
- r. Results of all stack tests and visible emission evaluations.
- s. Records of scheduled maintenance checks and readiness testing (Scheduled MCRT).
- t. Records of unscheduled maintenance, testing, and operator training.
- u. Records as required by Condition 29.

Compliance for the consecutive 12-month period in subsections above (as applicable) shall be demonstrated monthly by adding the total for the most recently completed month to the individual monthly totals for the preceding 11 months.

Compliance for the consecutive 365-day period in the subsections above (as applicable) shall be demonstrated daily by adding the total for the most recently completed day to the individual daily totals for the preceding 364 days.

These records shall be available for inspection by the DEQ and shall be current for the most recent five years, unless otherwise noted.
(9VAC5-80-1180 and 9VAC5-50-50) [11/6/2025]

NOTIFICATIONS

24. **Initial Notifications** – The permittee shall furnish written notification of the items below to the Regional Air Compliance Manager of the DEQ’s NRO at the following address:

Regional Air Compliance Manager
Department of Environmental Quality
13091 Crown Court
Woodbridge, VA 22193

The permittee shall submit one notification for each building or construction phase containing information on each emergency engine gen-set as described below:

- a. The actual date on which installation of the diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) in the building, or phase, commenced within 30 days after such date. The notification must contain the following:
 - i. Name and address of the permittee;
 - ii. The address of the affected source;
 - iii. Unit reference number and engine group of the initial unit installed; and

- iv. The date construction commenced.
- b. The date that the integration operational period started for each diesel engine gen-set (Engine Groups 1, 2A, 2B, 3A, 3B, 3C, 4, 5, and House Gen) within 15 days after the last generator at each building, or construction phase, completes its integration operational period. If a period of construction is paused or halted for 45 days this notification shall be provided to the DEQ within 15 days after completion of the integration operational period for the most recently installed diesel engine gen-set. The notification must contain the following:
 - i. Unit reference number and engine group;
 - ii. Engine information including make, model, engine family, serial number, model year, maximum engine power, engine displacement, fuel used;
 - iii. Installation date; and
 - iv. Integration operational period start and end dates.

For the purpose of this notification, the integration operational period is defined as: the period of time beginning with the first time the affect unit is started on-site and ending when the affected unit is fully integrated with the sources electrical system. In no case shall this period exceed 30 days.

(9VAC5-50-50 and 9VAC5-80-1180) [11/6/2025]

GENERAL CONDITIONS

25. **Permit Invalidation** – This permit to construct the emergency diesel engine gen-sets (Engine Groups 1, 2A, 2B, 3A, 3B, and House Gen) shall become invalid, unless an extension is granted by the DEQ, if:
- a. A program of continuous construction is not commenced within 18 months from the date of the permit amendment (November 19, 2024), or if
 - b. A program of construction is discontinued for a period of 18 months or more, or is not completed within a reasonable time, except for a DEQ approved period between phases of the phased construction of a new stationary source or project.

(9VAC5-80-1210)

26. **Permit Invalidation** – This permit to construct the diesel engine gen-sets (Engine Groups 3C, 4, and 5) shall become invalid, unless an extension is granted by the DEQ, if:

- a. A program of continuous construction is not commenced within 18 months from the date of this permit, or if
- b. A program of construction is discontinued for a period of 18 months or more, or is not completed within a reasonable time, except for a DEQ approved period between phases of the phased construction of a new stationary source or project.

(9VAC5-80-1210)

27. **Permit Suspension/Revocation** – This permit may be suspended or revoked if the permittee:

- a. Knowingly makes material misstatements in the permit application or any amendments to it;
- b. Fails to comply with the conditions of this permit;
- c. Fails to comply with any emission standards applicable to a permitted emissions unit;
- d. Causes emissions from the stationary source which result in violations of, or interfere with the attainment and maintenance of, any ambient air quality standard; or
- e. Fails to operate in conformance with any applicable control strategy, including any emission standards or emissions limitations, in the State Implementation Plan in effect at the time an application for this permit is submitted.

(9VAC5-80-1210 G)

28. **Right of Entry** – The permittee shall allow authorized local, state, and federal representatives, upon the presentation of credentials:

- a. To enter upon the permittee's premises on which the facility is located or in which any records are required to be kept under the terms and conditions of this permit;
- b. To have access to and copy at reasonable times any records required to be kept under the terms and conditions of this permit or the State Air Pollution Control Board Regulations;
- c. To inspect at reasonable times any facility, equipment, or process subject to the terms and conditions of this permit or the State Air Pollution Control Board Regulations; and

- d. To sample or test at reasonable times.

For purposes of this condition, the time for inspection shall be deemed reasonable during regular business hours or whenever the facility is in operation. Nothing contained herein shall make an inspection time unreasonable during an emergency.
(9VAC5-170-130 and 9VAC5-80-1180)

- 29. **Maintenance/Operating Procedures** – At all times, including periods of start-up, shutdown, and malfunction, the permittee shall, to the extent practicable, maintain and operate the affected source, including associated monitoring devices and air pollution control equipment, in a manner consistent with good air pollution control practices for minimizing emissions.

The permittee shall take the following measures in order to minimize the duration and frequency of excess emissions:

- a. Develop a maintenance schedule and maintain records of all scheduled and non-scheduled maintenance.
- b. Maintain an inventory of spare parts.
- c. Have available written operating procedures for equipment. These procedures shall be based on the manufacturer's recommendations, at a minimum.
- d. Train operators in the proper operation of all such equipment and familiarize the operators with the written operating procedures, prior to their first operation of such equipment. The permittee shall maintain records of the training provided including the names of trainees, the date of training and the nature of the training.

Records of maintenance and training shall be maintained on site for a period of five years and shall be made available to DEQ personnel upon request.
(9VAC5-50-20 E and 9VAC5-80-1180 D)

- 30. **Record of Malfunctions** – The permittee shall maintain records of the occurrence and duration of any bypass, malfunction, shutdown or failure of the facility or its associated air pollution control equipment that results in excess emissions for more than one hour. Records shall include the date, time, duration, description (emission unit, pollutant affected, cause), corrective action, preventive measures taken and name of person generating the record.
(9VAC5-20-180 J and 9VAC5-80-1180 D)

31. **Notification for Facility or Control Equipment Malfunction** – The permittee shall furnish notification to the Regional Air Compliance Manager of the DEQ’s NRO of malfunctions of the affected facility or related air pollution control equipment that may cause excess emissions for more than one hour. Such notification shall be made no later than four daytime business hours after the malfunction is discovered. The permittee shall provide a written statement giving all pertinent facts, including the estimated duration of the breakdown, within 14 days of discovery of the malfunction. When the condition causing the failure or malfunction has been corrected and the equipment is again in operation, the permittee shall notify the Regional Air Compliance Manager of the DEQ’s NRO.
(9VAC5-20-180 C and 9VAC5-80-1180)
32. **Violation of Ambient Air Quality Standard** – The permittee shall, upon request of the DEQ, reduce the level of operation or shut down a facility, as necessary to avoid violating any primary ambient air quality standard and shall not return to normal operation until such time as the ambient air quality standard will not be violated.
(9VAC5-20-180 I and 9VAC5-80-1180)
33. **Change of Ownership** – In the case of a transfer of ownership of a stationary source, the new owner shall abide by any current minor NSR permit issued to the previous owner. The new owner shall notify the DEQ’s NRO of the change of ownership within 30 days of the transfer.
(9VAC5-80-1240)
34. **Permit Copy** – The permittee shall keep a copy of this permit on the premises of the facility to which it applies.
(9VAC5-80-1180)

SOURCE TESTING REPORT FORMAT

Report Cover

1. Plant name and location
2. Units tested at source (indicate Ref. No. used by source in permit or registration)
3. Test Dates.
4. Tester; name, address and report date

Certification

1. Signed by team leader/certified observer (include certification date)
2. Signed by responsible company official
3. *Signed by reviewer

Copy of approved test protocol

Summary

1. Reason for testing
2. Test dates
3. Identification of unit tested & the maximum rated capacity
4. *For each emission unit, a table showing:
 - a. Operating rate
 - b. Test Methods
 - c. Pollutants tested
 - d. Test results for each run and the run average
 - e. Pollutant standard or limit
5. Summarized process and control equipment data for each run and the average, as required by the test protocol
6. A statement that test was conducted in accordance with the test protocol or identification & discussion of deviations, including the likely impact on results
7. Any other important information

Source Operation

1. Description of process and control devices
2. Process and control equipment flow diagram
3. Sampling port location and dimensioned cross section. Attached protocol includes: sketch of stack (elevation view) showing sampling port locations, upstream and downstream flow disturbances and their distances from ports; and a sketch of stack (plan view) showing sampling ports, ducts entering the stack and stack diameter or dimensions

Test Results

1. Detailed test results for each run
2. *Sample calculations
3. *Description of collected samples, to include audits when applicable

Appendix

1. *Raw production data
2. *Raw field data
3. *Laboratory reports
4. *Chain of custody records for lab samples
5. *Calibration procedures and results
6. Project participants and titles
7. Observers' names (industry and agency)
8. Related correspondence
9. Standard procedures

* Not applicable to visible emission evaluations